

Academic Qualifications

| Year           | Qualification    | Institute                             | CPI/%  |
|----------------|------------------|---------------------------------------|--------|
| 2022 - Present | B.Tech.          | Indian Institute of Technology Kanpur | 8.3/10 |
| 2022           | Class XII (CBSE) | Delhi Public School Nigahi            | 93.2%  |
| 2020           | Class X (CBSE)   | Delhi Public School Nigahi            | 95.2%  |

Scholastic Achievements

- Secured **Bronze medal** in **Inter-IIT Tech Meet 13.0 (2024)** for the Albatross Energetics' HVAC problem statement
- Awarded **KVPY Fellowship** for qualifying **KVPY SA 2020** with **All India Rank 641** amongst **1 lakhs+** candidates
- Secured a **Top 100** rank in **InMobi Codies 2026**, a flagship national-level hackathon organized by InMobi among IITs

Professional Experience

Software Engineering Intern | Samsara

(May'26-Jul'26)

|           |                                                                                                                           |
|-----------|---------------------------------------------------------------------------------------------------------------------------|
| Objective | Redesigned a <b>large-scale public API</b> 's pagination to fix unreliable behavior on large, sparse historical queries   |
| Approach  | Authored <b>RFCs</b> benchmarking <b>legacy, point-bounded &amp; DuckDB-based</b> designs to select the optimal approach  |
|           | Engineered a reusable <b>Go pagination engine</b> as a <b>point-counting wrapper</b> over a <b>byte-budgeted</b> reader   |
|           | Integrated pagination engine via <b>gRPC</b> across <b>microservices</b> while preserving <b>backward compatibility</b>   |
|           | Built <b>production observability</b> with Datadog dashboards & automated correctness monitors for staged rollout         |
|           | Hardened system with <b>table-driven, property-based, integration, and race testing</b> across varying workloads          |
| Outcome   | Reduced <b>p90 per-page latency</b> by <b>91.1%</b> versus legacy, validating zero errors across <b>200M+ data points</b> |

Key Projects

Deep Learning & Domain Adaptation | CHE629 Course Project | Prof. Sana Khanum 

(Jan'26-Apr'26)

- Reproduced **DeepCDA**, a compound-protein affinity model, with parallel **CNN-LSTM encoders** and two-sided attention layers
- Built a **config-driven** PyTorch pipeline with **SMILES/protein tokenization**, checkpointing, logging and reproducible seeding
- Reduced training cost **5x** via **epoch-level batch sampling** (20%), cutting compute under constrained GPU/CPU resources
- Extended the pipeline with **domain adaptation**, using adversarial **discriminator-encoder** training for feature alignment
- Validated the **reproduction** against classical and deep learning baselines, achieving **0.167 MSE** and **0.841 Concordance Index**

Event-Driven Kinetic Monte Carlo Engine | Undergraduate Project | Prof. Salman A. Khan

(Jan'25-Nov'25)

- Built a high-throughput, event-driven **Kinetic Monte Carlo engine** for scalable, stochastic lattice simulations in Python
- Implemented a **Binary Sum Tree**, reducing event selection and rate-update complexity from **O(N)** to **O(log N)** per step
- Evaluated **Monte Carlo convergence** and **sampling variance** using **confidence-interval** estimation across random seeds
- Built a **config-driven workflow** to automate simulation execution, statistical analysis, and visualization across parameter sweeps
- Improved the engine's measured event-processing throughput from **10** to over **100,000 events/second**, a **10<sup>4</sup>-fold** speedup

Parallel Computing | CS633 Course Project | Prof. Preeti Malakar 

(Feb'26-Apr'26)

- Parallelized a **3D stencil computation** using **MPI**, flattening sub-domains into padded **1D arrays** for cache locality
- Engineered **non-blocking halo exchanges** via **async messaging** and **null-processor routing** to avoid distributed deadlocks
- Profiled **scalability up to 96 processes** on **PARAM Rudra**, identifying **network communication** as the primary bottleneck

Binary Classification with Multi-Modal Features | CS771 Course Project | Prof. Piyush Rai 

(Oct'24-Nov'24)

- Evaluated **SVM, LogReg, Random Forest, XGBoost** and **LSTM** across datasets with distinct feature representations
- Applied **PCA, CountVectorizer, feature averaging**, and **one-hot encoding** for dataset-specific feature transformations
- Achieved highest individual model accuracy of **98.7%** using **flattened deep features** with optimized **logistic regression**

Model-Based & Policy-Gradient RL | EE675 Course Project | Prof. Washim Mondal 

(Jan'26-Apr'26)

- Implemented **Value Iteration and Policy Iteration** using sparse transition kernels, scaling to **390K+** states efficiently
- Estimated unknown transition dynamics via **Monte Carlo sampling**, quantifying  $L_1$  value-function error across sampling budgets
- Trained a **PyTorch REINFORCE policy network**, evaluating convergence & stability across **48** hyperparameter configurations

Technical Skills

- Programming Languages & Utilities:** C, C++, Go, Python, TypeScript, JavaScript, SQL, Bash, GLSL, LaTeX
- Software & Libraries:** Git, GitHub, Docker, Postman, gRPC, MPI, XGBoost, Next.js, NestJS, PostgreSQL, Prisma, Numba

Relevant Coursework

\* - Ongoing

|                                 |                                  |                             |
|---------------------------------|----------------------------------|-----------------------------|
| Data Structures and Algorithms  | Introduction to Machine Learning | Fundamentals of Computing   |
| Intro to Reinforcement Learning | AI in System Biology             | Signals, Systems & Networks |
| Parallel Computing              | Computer Arithmetic on FPGA*     | Convex Optimization in ML   |

Leadership Experience

Coordinator | Center for Mental Health and Wellbeing, IIT Kanpur

(Apr'25-Apr'26)

- Spearheaded **3-tier** team, overseeing **22** Core Team Members, **350+** Guides/Mentors; led **9-day** Orientation for **1200+** students
- Managed a **INR 40L+** budget and secured **INR 10L** in sponsorships through strategic vendor and bank partnerships
- Special Invitee to **Senate, IITK**, institute's highest decision & policy making body and **Student Wellness Committee**, IITK
- Revamped the **CMHW website** (**183%** engagement increase) and expanded **online counselling** access, driving a **145%** rise